


# CYLSEC Virtual Lab Environment Setup Guideline

**UTM Installation and Configuration for Kali Linux and Metasploitable 2 for macOS M-series Chips**

**Author:** Dan Gabriel B. Romulo

**Date Created:** July 19, 2026

# 1. Introduction

## 1.1 Purpose

This guideline instructs CSB students and independent users with macOS systems that use M-series chips how to set up UTM as an alternative to VirtualBox. Following it ensures every user ends up with an identical, reproducible environment: one attacker machine (**Kali Linux**) and one intentionally vulnerable target (**Metasploitable 2**).

## 1.2 Scope

This document covers:

- Installing UTM on a Mac
- Deploying Kali Linux and Metasploitable 2 as guest VMs
- Basic network isolation for the lab
- Validation, troubleshooting, and safe-handling guidance

## 1.3 Intended Audience

CYLSEC trainees, stakeholders and CSB students building a local practice lab for penetration testing, tool familiarization, or internal exercises. Basic command-line comfort is assumed.

### 1.4 Definitions and Abbreviations

| Term | Meaning |
|---|---|
| UTM | A free, QEMU-based virtualization app for macOS/iOS |
| VM | Virtual Machine |
| ISO | Optical disc image file, commonly used to install an OS |
| ARM64 / Apple Silicon | CPU architecture used by M-series Macs (M1–M4) |
| x86_64 / amd64 | CPU architecture used by Intel Macs, and by Metasploitable 2 and most legacy Linux builds |
| Virtualize (UTM) | Runs a guest OS natively — guest architecture must match host architecture |
| Emulate (UTM) | Runs a guest OS of a different architecture than the host, via software emulation (slower) |
| QEMU | The open-source emulator/virtualizer UTM is built on |
| qemu-img | Command-line tool (installed via Homebrew) used to convert disk image formats |
| VMDK | VMware's native virtual disk format |
| QCOW2 | QEMU's native virtual disk format |
| NAT / Shared Network | A VM network mode where the VM shares the host's internet connection but is hidden from the physical LAN |
| Bridged | A VM network mode where the VM appears as its own device on the physical network |
| Clone (UTM) | An instant, full duplicate of a VM — UTM's substitute for a "snapshot," since UTM has no built-in snapshot/restore feature |

# 2. Prerequisites

## 2.1 Hardware Requirements

- A Mac with Apple Silicon (M1–M4)
- 8 GB RAM is workable for running both VMs together; 16 GB is recommended for a smoother experience
- ~75 GB free disk space, broken down roughly as:
  - – Kali Linux: 40–60 GB
  - – Metasploitable 2: 10–15 GB

## 2.2 Software Requirements

- A currently supported version of macOS
- UTM (latest stable release)

- Homebrew (to install qemu-img for converting the Metasploitable 2 disk image)

### 2.3 Required Files and Legal Sourcing

| VM | Official Source | Notes |
|---|---|---|
| Kali Linux | https://www.kali.org/get-kali/#kali-installer-images | Choose Apple Silicon (ARM64) for M-series Macs, or the standard amd64 installer for Intel Macs |
| Metasploitable 2 | https://sourceforge.net/projects/metasploitable/ | Ships as a VMware disk image; will need format conversion (Chapter 5) |

# 3. Installing UTM

### 3.1 Download Source

https://mac.getutm.app/

### 3.2 Installation Procedure

Download the .dmg file.

1. Open it, then drag **UTM.app** into your **Applications** folder.
2. Launch UTM from Applications. If macOS Gatekeeper blocks the app on first launch, go to **System Settings → Privacy & Security** and click **Open Anyway**.

### 3.3 Interface Overview

The main window lists all your VMs.

- The **"+"** button opens the **Create a New Virtual Machine** wizard.
- Each VM has a **Settings** (gear icon) for hardware/network/drive configuration, and **Play/Stop** controls.
- **Browse UTM Gallery** provides pre-built VM templates for some operating systems (not used for this guide, since we're building each VM manually for consistency).

# 4. Deploying the Kali Linux Virtual Machine

### 4.1 Obtaining the Kali Linux Image

Go to the Kali Linux Get Kali page and open **Installer Images**.

**Apple Silicon Mac:** download the **Apple Silicon (ARM64)** installer ISO.

**Intel Mac:** download the standard **amd64** installer ISO.




## 4.2 Creating the VM in UTM

Click **"+"** → **Create a New Virtual Machine**.

Choose **Virtualize** (matches host architecture on both Apple Silicon and Intel Macs, since we downloaded the matching Kali image above).

Select **Linux**.

Under **Boot ISO Image**, browse to the Kali ISO you downloaded.

Hardware: allocate **4 GB+ RAM**, **2+ CPU cores**; enable hardware OpenGL acceleration if offered, for a smoother desktop experience.

Storage: **40–60 GB**.

Shared Directory: optional — skip for now, can be configured later from the VM toolbar.

On the Summary screen, rename the VM to **"Kali Linux"** and click **Save**.

## 4.3 Installation Walkthrough

Start the VM and step through the Debian-based installer: language, location, keyboard layout.

Set hostname (e.g., `kali`) and leave domain blank unless your lab requires one.

Set a **root password** (or configure a sudo-capable user, per current Kali installer defaults) — do not reuse any personal or production password here.

Disk partitioning: choose **Guided – use entire disk**, and **All files in one partition** (simplest, appropriate for a lab VM).

Confirm and write changes to disk.

When asked about a network mirror, accept the default (recommended, so `apt` works out of the box).

Software selection: choose the **Xfce** desktop environment (lightweight, fast under virtualization) plus the **top10** tools collection and **default** tools.

Install **GRUB** to the primary disk when prompted.

Reboot when installation completes; remove/detach the ISO from the virtual CD/DVD drive so it doesn't boot into the installer again (VM Settings → Drives → CD/DVD → Clear).

Go to Edit > + New > Serial > Save

This will make it so when you boot it for installation, another window will pop up and act as the controller.

Go to Network > Network Mode > change to Bridged (Advanced)

### 4.4 Post-Installation Setup

Log in with the account created during installation.

1. Update the system:

```
sudo apt update && sudo apt full-upgrade -y
```

# 5. Deploying the Metasploitable 2 Virtual Machine

### 5.1 Obtaining Metasploitable 2

Download `metasploitable-linux-2.0.0.zip` (~870 MB) from the official SourceForge project page (or the mirrored link on Rapid7's Metasploit documentation site).

1. Extract it:

```
cd ~/Downloads
unzip metasploitable-linux-2.0.0.zip
```

This produces a `Metasploitable.vmdk` file (plus VMware config files you won't need).

### 5.2 Importing the VM into UTM

UTM/QEMU works with `.qcow2`, not `.vmdk`, so the disk needs converting.

1. Install Homebrew in your terminal, if you haven't already (see brew.sh for the official install command).
2. Install QEMU's command-line tools:

```
brew install qemu
```

3. Convert the disk:

```
cd ~/Downloads/Metasploitable2-Linux
qemu-img convert -f vmdk -O qcow2 Metasploitable.vmdk Metasploitable2.qcow2
```

This can take a few minutes.

### 5.3 Creating the VM and Attaching the Disk

Click **"+" → Create a New Virtual Machine**.

Choose **Emulate** (required on Apple Silicon since this is an x86_64-only image; Intel Mac users may also use Emulate here for simplicity, since we're importing an existing disk rather than booting an ISO).

Select **Other** as the operating system.

When asked about a boot device, choose **None** / leave ISO empty (skip ISO boot) — we'll attach the converted disk directly.

Hardware: **512 MB–1 GB RAM** is enough (Metasploitable is intentionally lightweight); leave CPU cores at default.

Storage: this can be minimal, since we will delete the auto-created drive and import our own.

On the Summary screen, check **Open VM Settings**, name the VM **"Metasploitable 2"**, and save.

In **VM Settings → Drives**, remove the empty drive UTM created by default.

Click **New → Import**, and select the `Metasploitable2.qcow2` file you created in step 5.2.

In **VM Settings → QEMU**, disable **UEFI Boot** (Metasploitable expects legacy BIOS boot).

Configure networking per Chapter 6 before first boot.

### 5.4 First Boot and Login

Start the VM. It will boot to a console login prompt. Log in using the vendor-published default credentials for this training image: `msfadmin` / `msfadmin` (this is intentionally published by the vendor as the standard login for this deliberately vulnerable practice VM — not a security issue to document here).

### 5.5 Safety Notes

This machine is intentionally riddled with vulnerabilities for training purposes. It must **never** be connected to a production network, the open internet, or any network shared with untrusted devices. See Chapter 8 for mandatory isolation practices.

## 6. Network Configuration for the Lab

### 6.1 UTM Network Modes Overview

Each VM's network adapter can be set (in VM Settings → Network) to one of several modes:

- **Shared Network** — the default mode. The VM gets outbound internet access through the host's connection, but is hidden from the physical LAN behind UTM's built-in NAT.
- **Bridged (Advanced)** — the VM appears as its own device directly on the physical network, with an IP address from your router. Not recommended for lab VMs, and never recommended for Metasploitable 2.

- **Host Only** — places the VM on a network isolated from the physical network. Behavior for VM-to-VM visibility in this mode should be verified directly (Section 6.3) rather than assumed, since it can vary by UTM version.

### 6.2 Recommended Lab Topology

For this lab, set both Kali Linux and Metasploitable 2 to **"Shared Network"** mode. This keeps both VMs behind UTM's NAT — invisible to your campus/office network or the wider internet — while still allowing them to reach each other and, if needed, to reach the internet for updates.

If a given exercise requires the lab to have **zero** internet access, the simplest reliable approach is to physically disconnect the Mac's Wi-Fi/Ethernet before starting the VMs, rather than relying on a specific UTM network mode to guarantee full isolation.

### 6.3 Verifying inter-VM Connectivity

Note the IP address assigned to each VM (`ip a` on both Linux guests).

From the Kali VM, ping the Metasploitable 2 IP address to confirm connectivity.

OR

Run `arp -an` in both terminals and check that both IP addresses show; if so, the two VMs are connected.

Confirm that devices on your actual home/campus network **cannot** ping or reach these VMs — this is the isolation check that matters most.

# 7. Troubleshooting Common Issues

| Issue | Likely Cause / Fix |
|---|---|
| Kali installer screen goes black shortly after boot (Apple Silicon) | Known UTM display bug during graphical install on some versions. Add a Serial device (VM Settings → System → New → Serial) and install in console-only mode. |
| Mouse cursor "stuck" inside a VM window | Press Control + Option together, or use the mouse-capture icon in the VM toolbar, to release/recapture the pointer. |
| qemu-img: command not found | Homebrew/QEMU isn't installed. Run `brew install qemu`. |
| No Wi-Fi adapter visible inside a Linux guest | Expected — UTM presents the host's network (Wi-Fi or Ethernet) to the guest as a single wired adapter. |
| Metasploitable 2 (x86_64 on Apple Silicon) is slow to boot or respond | Expected — software emulation is slower than native virtualization. Be patient during boot, and give the VM a minute or two after the login prompt appears before running scans. |
| VM boots straight back into the installer after finishing setup | The ISO is still attached as the boot device. Go to VM Settings → Drives → CD/DVD and clear it. |
| Network settings don't seem to apply after changing modes | Restart the affected VM after changing network mode; live changes aren't always picked up immediately. |

# 8. Lab Safety and Handling Guidelines

## 8.1 Network Isolation

Both lab VMs should remain on an isolated Shared Network as described in Chapter 6. **Never** set Metasploitable 2 to Bridged mode, and never expose it — or any lab VM — to a production, home, or campus network.

## 8.2 Credential and Data Handling

Never enter real personal credentials, CYLSEC production credentials, or client data into any lab VM. Treat `msfadmin/msfadmin` and any Kali account password as lab-only, disposable credentials.

## 8.3 Snapshot and Rollback Practice

UTM has no built-in snapshots, so use **Clone** instead: in the UTM VM list, right-click the VM and choose **Clone** (the VM must be shut down first).

Clone each VM immediately after a clean, fully configured install, and again before any exercise that will meaningfully alter system state (e.g., successful exploitation, privilege escalation, persistence testing). This keeps the lab reproducible for every trainee.

## 8.4 Responsible Use Policy

Reference CYLSEC's acceptable-use/ethics policy for training environments here. All techniques practiced in this lab are for authorized, defensive, and educational purposes only, against machines the trainee owns or is explicitly authorized to test.